

Project Proposal

“Windsor” Integrated Management System — Proposal and Software Requirements Specification

Change History

Table 1. Change history

ID	Version	Date	Decision	Reason	Affected sections
D-01	v1	2026-10-01	Add requirement RF-CLI-004 “Hall booking view” to Client Management.	v0 had no requirement giving the Client a view to book a hall; RF-SAL-003 only records bookings.	Project Scope; Software Requirements Specification
D-02	v1	2026-10-01	Schedule the booking view and its public access in Sprints 3–4. Timeline, milestones, and team stay unchanged.	Sprints 3–4 carry 7 requirements against 9 in the neighboring sprint pairs, and Event Halls is delivered in Sprints 1–2.	Agile Approach and Schedule
D-03	v1	2026-10-01	Publish the system through a public Nginx proxy on a Vultr VPS, linked to the client’s on-premises LAN by a VPN tunnel.	v0 assumed a LAN-only deployment, and Clients need to book over the internet. Chosen from 12 alternatives for the lowest effort and cost.	Proposed Technologies; Software Requirements Specification
D-04	v1	2026-10-01	Add the third-party costs of the Vultr VPS with DDoS protection and of a domain name.	D-03 adds recurring costs. The budget of \$1,200,000 MXN stays unchanged.	Project Scope
D-05	v1	2026-10-01	Add the risks of internet exposure and of depending on the hall’s internet link, and extend the third-party dependency to Vultr.	Consequences of D-03.	Identified Risks
D-06	v1	2026-10-02	Name the exact Vultr server (Cloud Compute Regular Performance <code>vc2-1c-1gb</code> , Mexico City) and break down its costs.	D-03 and D-04 gave only the provider and a total, which is not enough to contract or audit the service.	Project Scope; Proposed Technologies; Software Requirements Specification
D-07	v1	2026-10-02	Add two communication diagrams: deployment communication and the module messages of the booking flow.	The architecture diagram does not show the VPN tunnel or how the modules interact when a Client books a hall.	Software Requirements Specification

ID	Version	Date	Decision	Reason	Affected sections
D-08	v1	2026-10-02	Add the cost of SendGrid (Essentials plan) to the third-party costs.	SendGrid was already part of the architecture, but its cost was not budgeted and it no longer offers a permanent free plan.	Project Scope; Identified Risks
D-09	v1	2026-10-02	Redraw the architecture diagram at a larger size, with the same layers and components. Correct the spelling of its labels and remove the connector that joined the application layer directly to the presentation layer.	The original image was hard to read at page size, and that connector contradicted RNF-SEG-001.	Software Requirements Specification
D-10	v1	2026-10-02	Add a timeline of the sprint schedule. The sprint table stays unchanged.	A timeline shows at a glance the order of the increments and where the milestones fall.	Agile Approach and Schedule
D-11	v1	2026-10-02	Register the domain <code>saloneswindsor.mx</code> with Porkbun and use its real price, its included SSL certificate, and its DNS.	The domain cost was an estimate, and the certificate and DNS provider were not defined.	Project Scope; Identified Risks; Software Requirements Specification
D-12	v1	2026-10-02	Remove the Assumptions and Notes section.	Removed at the author's request.	Assumptions and Notes
D-13	v1	2026-10-02	Describe the client as one event venue with six halls in a single facility that share one LAN, instead of a chain of venues.	The halls are all in the same facility. "Chain" suggested several sites, which would put the LAN-based deployment in doubt.	General Information; Justification and Objective; Software Requirements Specification

General Information

Table 2. General information

Field	Detail
Project name	Windsor — Integrated Management System
Development company	General México Software Co.
Client	Event venue with six halls in a single facility, in the Monterrey metropolitan area
Type	Custom software development (web application)
Budget	\$1,200,000 MXN
Timeline	6 months to delivery date
Document version	v1

Justification and Objective

The client operates an established event venue (weddings, quinceañeras, corporate events) with six halls in a single facility and catering service. It currently lacks a unified system to run its operations. The project’s objective is to design, build and deliver a custom integrated management system that centralizes the business —from event booking and supplier services to payments and QR access control— on a single web platform.

Windsor is a web application for booking and managing event halls. It connects clients, service suppliers, administrators, and security staff on a single platform to reserve venues, hire complementary services, manage guest lists, and control access at events through QR codes.

Project Scope

The project covers seven modules:

1. Authentication and Access Control — registration, login, roles, and password recovery
2. Event Halls — hall catalog, availability, bookings, and maintenance blocking
3. Client Management — client profile, hall booking view, event history, and event management panel
4. Suppliers and Services — supplier service catalog, schedule, and ratings
5. Purchases and Payments — cost calculation, purchase orders, payment status, and receipts
6. Guests and QR Access Control — guest lists, QR generation, and validation at the door
7. Email Notifications — confirmations, QR invitations, schedule changes, and alerts

Each module is described in detail, with its requirements, in the Software Requirements Specification.

Third-Party Software

Employee management and electronic invoicing (CFDI) are not developed in this project. Invoicing is handled with Facturama, a third-party CFDI service contracted on its annual plan, so the team focuses on the seven modules above. The purchase receipts in RF-COMP-004 are purchase summaries; official invoices are issued through Facturama.

Clients reach the system over the internet through a public proxy hosted on a Vultr virtual private server (VPS), with DDoS protection, under the domain `saloneswindsor.mx`, registered with Porkbun. The server is a Vultr Cloud Compute Regular Performance instance, plan `vc2-1c-1gb`, in the Mexico City region. Email delivery uses the SendGrid Essentials plan, which covers up to 50,000 emails per month. **Table 3** lists the third-party costs, which are covered by the project budget of \$1,200,000 MXN, and **Table 4** breaks down the Vultr cost.

Table 3. Third-party usage costs (MXN)

Service	Use in the project	Plan	Annual cost
Facturama	Electronic invoicing (CFDI) for event purchases	Annual plan	\$1,650 MXN
Vultr	Public Nginx proxy on a Cloud Compute Regular Performance <code>vc2-1c-1gb</code> instance in Mexico City, with DDoS protection	Monthly plan, \$15 USD per month	\$3,291 MXN (\$180 USD)
SendGrid	Email delivery over SMTP: confirmations, QR invitations, and alerts	Essentials plan, \$19.95 USD per month	\$4,377 MXN (\$239.40 USD)
Porkbun	Domain <code>saloneswindsor.mx</code> , with its DNS and SSL certificate included	Annual registration: \$35.57 USD the first year, then \$41.23 USD	\$650 MXN (\$35.57 USD)
Total, first year			\$9,968 MXN

From the second year the domain renews at \$754 MXN (\$41.23 USD), so the annual total becomes \$10,072 MXN. The domain uses Porkbun's DNS, and Porkbun issues the SSL certificate that enables HTTPS at no cost while the domain stays on that DNS.

Table 4. Vultr cost breakdown

Item	Detail	Monthly cost	Annual cost
Instance	Cloud Compute Regular Performance <code>vc2-1c-1gb</code> , Mexico City: 1 vCPU, 1 GB RAM, 25 GB SSD, and 1 TB of transfer per month	\$5 USD	\$1,097 MXN (\$60 USD)
DDoS protection	10 Gbps of mitigation on the public IP address	\$10 USD	\$2,194 MXN (\$120 USD)
Included	One public IPv4 address, Vultr Firewall, and DNS	\$0 USD	\$0 MXN
Total contracted		\$15 USD	\$3,291 MXN (\$180 USD)
Automatic backups	Not contracted: the proxy stores no data and its configuration is kept in the repository	\$1 USD	—
Extra transfer	Not contracted: \$0.01 USD per GB above 1 TB per month	—	—
Additional IPv4	Not contracted: one address is enough	\$3 USD	—
Snapshots	Not contracted: \$0.05 USD per GB per month	—	—

The VPN tunnel (WireGuard) has no cost, and the SSL certificate comes with the Porkbun domain. Prices are before taxes: if Vultr charges the 16% IVA in Mexico, the Vultr total rises by \$527 MXN per year.

Project Team and Roles

Table 5. Project team and roles

Member	Role and skills	Amount (MXN)
General Director	Owner / founder; developer and client-relationship lead	N/A (owner)
Project Manager	Project management and UI/UX design (Figma)	\$24,000 MXN
Senior Developer (lead)	Fullstack technical lead; builds the pipeline and coordinates support. Angular, NestJS, TypeScript, PostgreSQL, Docker	\$23,000 MXN
Mid-level Developer	Fullstack module development (on-site). NestJS, Angular, PostgreSQL	\$16,000 MXN
Junior Developer	Module development and support (remote). Angular, TypeScript, Node	\$14,000 MXN
Monthly team cost	Sum of fixed monthly salaries (excludes the General Director)	\$77,000 MXN
Annual team cost	Monthly team cost × 12	\$924,000 MXN
Total project team cost	Monthly team cost × 6 months	\$462,000 MXN
25% profit	25% of the total project team cost	\$115,500 MXN
Total with 25% profit	Total project team cost + 25% profit	\$577,500 MXN

Proposed Technologies

- **Backend:** TypeScript / NestJS
- **Frontend:** Angular
- **Database:** PostgreSQL
- **Infrastructure:** Docker on the client's on-premises servers (LAN), published through a public Nginx proxy on a Vultr VPS (Cloud Compute `vc2-1c-1gb`, Mexico City) over a VPN tunnel
- **Internal automation:** a CLI tool (Python) to scaffold backend and frontend and seed the database, reducing repetitive boilerplate.

Deliverables

- Functional web system with the 7 in-scope modules.
- Modeled and seeded database.
- Development and deployment pipeline (containers).
- Basic technical documentation of the solution.

Agile Approach and Schedule

The project runs on Scrum: 2-week sprints with incremental delivery. Each sprint closes with a Sprint Review with the client to validate and re-prioritize the Product Backlog; per-sprint ceremonies: planning, daily, review and retrospective. Progress is estimated in user stories and tracked by team velocity.

Table 6. Sprint schedule and milestones

Sprint (2 wks each)	Sprint goal (increment)	Milestone
Sprint 0 (wk 1–2)	Base architecture, CI/CD pipeline, data model and initial backlog	—
Sprints 1–2 (wk 3–6)	Authentication and Access Control; Event Halls	—
Sprints 3–4 (wk 7–10)	Client Management, including the hall booking view and its public access (Vultr proxy and VPN tunnel); Suppliers and Services	—
Sprints 5–6 (wk 11–14)	Purchases and Payments; Guests and QR Access Control	MVP release
Sprints 7–8 (wk 15–18)	Email Notifications and cross-module integration	—
Sprints 9–12 (wk 19–26)	Hardening, testing, UAT, training and delivery	Final release

Figure 1 shows the same plan as a timeline. The baseline sets no calendar start date, so the timeline is expressed in weeks.

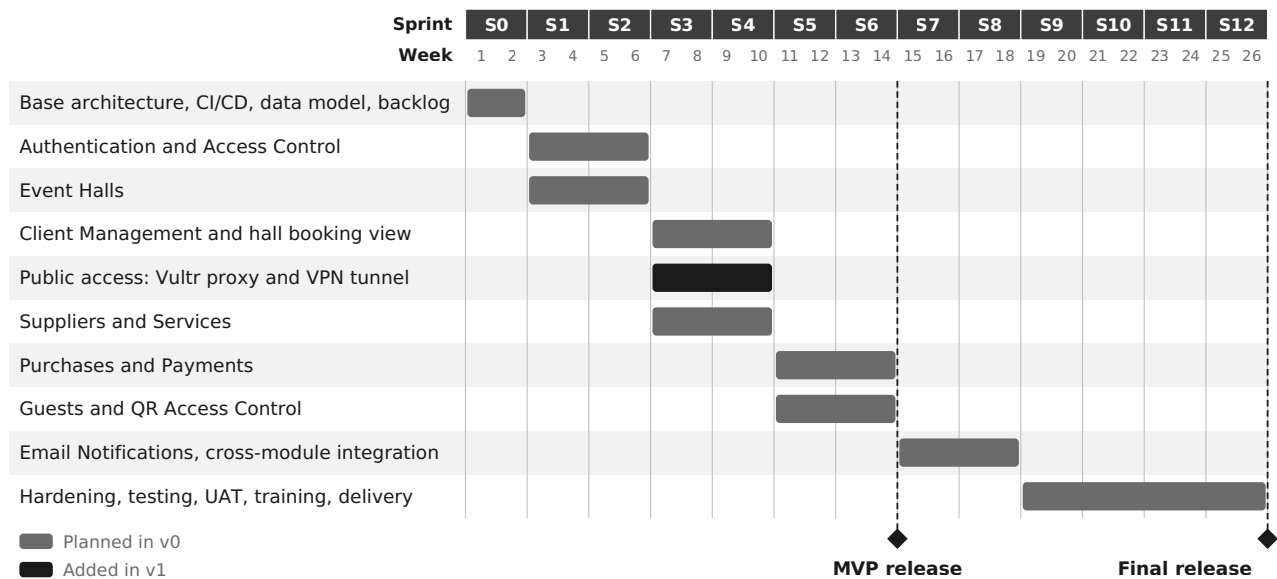


Figure 1. Sprint schedule timeline and milestones

Definition of Done: reviewed code (PR), unit and integration tests, and the increment deployed to a QA environment.

Scrum roles: Product Owner (prioritizes the backlog and validates with the client), Scrum Master (facilitates and removes impediments) and the Development Team.

Identified Risks

Table 7. Identified risks

Risk	Description	Impact
Key-person dependency	The project rests on a dual role (coordinator + lead developer) with minimal support.	High
Resource reassignment	Staff can be moved to other group work without change control.	High
PM authority	The PM tracks status but does not decide on resources; the owner does.	Med-high
Unprotected scope	No formal change-control process over scope.	High
Client concentration	A delay from another client can drain this project's resources.	Med-high
Third-party dependency	Invoicing depends on Facturama, email delivery on SendGrid, and public access on Vultr and Porkbun; their prices or availability can change.	Medium
Internet exposure	The system is reachable from the internet through the public proxy, which widens the attack surface.	Med-high
Hall internet link	Online booking depends on the hall's internet link. If it fails, Clients cannot reach the system, although on-site operation continues on the LAN.	Medium

Software Requirements Specification

Module Overview

Authentication and Access Control

This module manages user registration and login for clients and suppliers, using JWT tokens to keep sessions secure. It also restricts what each user can see and do based on their role (Administrator, Client, Supplier, or Security Guard), and lets users recover their password through a temporary token sent by email.

Event Halls

This module lets the Administrator register event halls with their name, location, maximum capacity, and hourly cost, upload photos of each hall, and check availability by date and time range. It also keeps a booking schedule that prevents overlapping or duplicate reservations, and allows the Administrator to block dates or time slots for maintenance or private events.

Client Management

This module gives clients a profile where they can view and update their personal data, contact information, and notification preferences. It also gives clients a booking view where they browse the hall catalog, check a hall's availability by date and time range, and book an available slot. Clients can also see a detailed history of their past, active, and cancelled events with invoices or summaries, and manage their guest list and hired supplier services from a dedicated panel for each active event.

Suppliers and Services

This module allows suppliers to register and update the services they offer (catering, entertainment, music, decoration, etc.) along with descriptions and prices, and to attach one or more of these services to a hall reservation. Suppliers can view a calendar with the dates, times, and halls where they must deliver their services, and clients can rate and review a supplier once an event has concluded.

Purchases and Payments

This module calculates the total cost of an event by adding the hall's hourly cost to the accumulated cost of the selected supplier services. It generates and stores a purchase order linked to the client and the event, allows the Administrator or the payment gateway to update the payment status (Pending, Completed, Refunded), and produces a detailed digital receipt that the client can view or download.

Guests and QR Access Control

This module lets clients add guests one by one or through a bulk upload with contact details, and automatically generates a unique QR code for each guest and event. Security guards use the web application to scan and validate each QR code at the door; once a code is successfully scanned it is marked as "Used" so it cannot be reused, and clients can resend a guest's QR code individually if it gets lost.

Email Notifications

This module automatically emails clients to confirm their hall booking and hired services, sends each registered guest their QR-coded invitation, and notifies clients, suppliers, and guests by email whenever an event's date changes or the event is cancelled. It also alerts the Administrator when a new supplier requests registration or when an external integration fails.

Functional Requirements

Table 8. Functional requirements

ID	Requirement	Description
RF-AUTH-001	User registration	The system must allow clients and suppliers to register by entering their basic information and access credentials.
RF-AUTH-002	JWT authentication	The system must validate user credentials and generate a JWT token to manage secure sessions.
RF-AUTH-003	Role-based access control	The system must restrict views and actions according to the assigned role (Administrator, Client, Supplier, Security Guard).
RF-AUTH-004	Password recovery	The system must allow users to recover access through a temporary token sent to their registered email.
RF-SAL-001	Event hall registration	The system must allow the Administrator to register halls, defining name, location, maximum capacity, and hourly cost.
RF-SAL-002	Hourly availability lookup	The system must allow checking a hall's availability on any date of the year, filtering by specific time ranges.
RF-SAL-003	Booking schedule	The system must record bookings by time range, preventing overlap or duplicate schedules for the same hall.
RF-SAL-004	Hall image upload	The system must allow uploading and storing photographs of the halls.
RF-SAL-005	Date blocking for maintenance	The system must allow the Administrator to disable halls or specific time ranges on given days for maintenance or private events.
RF-CLI-001	Client profile management	The system must allow the Client to view and update their personal data, contact information, and notification methods.
RF-CLI-002	Event history	The system must show the Client a detailed list of their past, active, and cancelled events with their respective invoices or summaries.
RF-CLI-003	Event management panel	The system must provide the Client with a panel to manage the guest list and the hired supplier services for an active event.
RF-CLI-004	Hall booking view	The system must provide the Client with a view to browse the hall catalog, check a hall's availability by date and time range, and book an available slot.
RF-PROV-001	Supplier service registration	The system must allow Suppliers to register and update their services (catering, entertainment, music, decoration) with descriptions and prices.
RF-PROV-002	Linking services to events	The system must allow attaching one or more supplier services to an event hall booking.

ID	Requirement	Description
RF-PROV-003	Supplier schedule lookup	The system must offer Suppliers a calendar to view the dates, times, and halls where they must deliver their services.
RF-PROV-004	Supplier rating	The system must allow the Client to rate and leave a review of the services provided by the Supplier once the event has concluded.
RF-COMP-001	Total purchase cost calculation	The system must calculate the total amount by adding the hourly hall cost and the accumulated cost of the selected supplier services.
RF-COMP-002	Purchase order registration	The system must generate and store the purchase order in pending or paid status, associating it with the client and the event.
RF-COMP-003	Transaction status change	The system must allow the Administrator or the payment gateway to update the payment status (Pending, Completed, Refunded).
RF-COMP-004	Purchase receipt generation	The system must generate a detailed summary of the purchase in digital format for the client to view or download.
RF-INV-001	Guest list upload	The system must allow the Client to enter guests individually or perform a bulk upload with contact details.
RF-INV-002	Unique QR code generation	The system must generate a unique hash and represent it graphically as an exclusive QR code for each guest and registered event.
RF-INV-003	QR scanning and validation	The system must allow the Security Guard to validate the authenticity of the QR code presented by the guest through the web application.
RF-INV-004	Real-time access control	The system must mark the QR code as “Used” after the first successful scan to prevent the same entry from being reused.
RF-INV-005	QR code resend	The system must allow the Client to individually resend the QR-coded ticket if a guest loses it.
RF-MAIL-001	Purchase confirmation notification	The system must send an automatic email to the Client confirming the hall booking and the hired services.
RF-MAIL-002	Bulk QR access delivery	The system must automatically send invitations with the packaged QR code to the email address of each registered guest.
RF-MAIL-003	Schedule change notification	The system must notify Clients, Suppliers, and Guests by email of any change in date or cancellation of the event.
RF-MAIL-004	Administration alerts	The system must notify the Administrator when a new Supplier requests registration or when a failure occurs in external integrations.

Non-Functional Requirements

Table 9. Non-functional requirements

ID	Requirement	Description
RNF-SEG-001	Reverse proxy concealment	The NestJS monolithic backend must operate behind a Proxy server (Nginx) without publicly exposing its private IP or ports.
RNF-SEG-002	Database isolation	The database must be hosted on a physically independent server with no public access, accepting connections only from the backend's IP.
RNF-PER-001	API response time	Requests processed by the NestJS routes must respond with an average time under 300 ms under normal traffic conditions.
RNF-USAB-001	Responsive interface	The Angular web application must adapt its layout to desktop browsers and mobile devices to facilitate QR reading at the door.
RNF-DISP-001	Service availability	The system must maintain 99.5% operational availability throughout the year to guarantee real-time queries and validations.
RNF-COMP-001	Scanning compatibility	The QR validation module must be compatible with the cameras of mobile devices running modern browsers (Chrome, Safari).

Architecture

Figure 2 shows the architecture of the party hall management system, and Table 10 summarizes its layers.

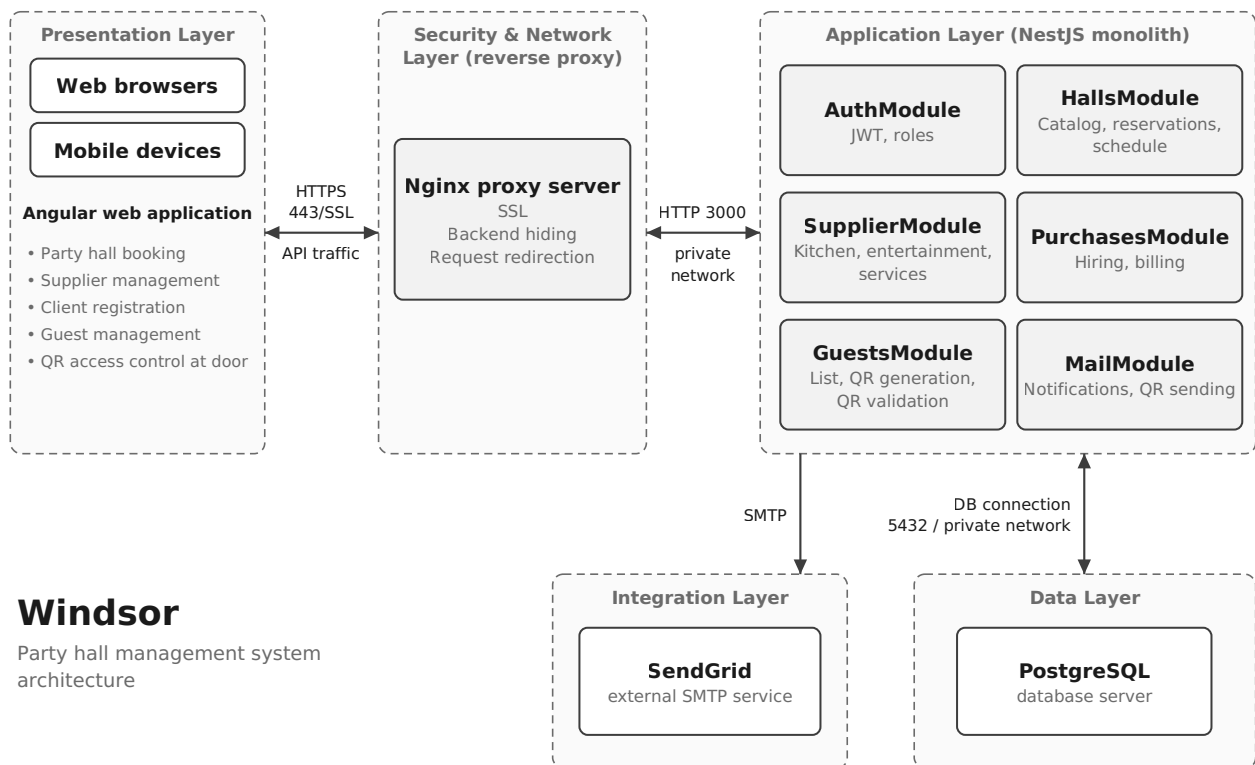


Figure 2. Windsor: party hall management system architecture

Table 10. Architecture layers

Layer	Components	Responsibilities and connections
Presentation	Web browsers, mobile devices	Party hall booking, supplier management, client registration, guest management, and QR access control at the door. Sends API traffic over HTTPS (443/SSL).
Security & Network	Nginx proxy server (reverse proxy) on a Vultr VPS (vc2-1c-1gb , Mexico City); VPN tunnel to the client's LAN	SSL with the certificate issued by Porkbun, backend hiding, DDoS protection, and request redirection. Forwards traffic over HTTP (3000 / private network) through the VPN tunnel.
Application	NestJS monolith: AuthModule, HallsModule, SupplierModule, PurchasesModule, GuestsModule, MailModule	JWT and roles; catalog, reservations, and schedule; supplier services (kitchen, entertainment, and services); hiring and billing; guest list, QR generation, and QR validation; notifications and QR sending.
Integration	External SMTP service (SendGrid); external payment gateway (vendor to be defined); electronic invoicing service (Facturama)	Email delivery over SMTP; payment status updates for purchase orders (RF-COMP-003); official invoices for purchases.
Data	Database server (PostgreSQL); file storage	Persistence over a DB connection (5432 / private network). Hall photographs (RF-SAL-004) are stored by the application in file storage, with their references in PostgreSQL.

Deployment: the services run in Docker containers, and the CI/CD pipeline deploys each increment to a QA environment before delivery, as required by the Definition of Done in Agile Approach and Schedule. The NestJS monolith and PostgreSQL run on the client's on-premises servers (LAN). The six halls are in the same facility and share that LAN. The public Nginx proxy runs on a Vultr VPS, and the LAN opens an outbound VPN tunnel to it, so no inbound port is opened on the client's network; this tunnel is the private network between the proxy and the monolith in **Figure 2**. On-site staff and security guards keep using the system over the LAN, so hall operation does not depend on the internet link.

Communication

Figure 3 shows how each party reaches the system. Clients resolve `saloneswindsor.mx` through Porkbun's DNS and then connect over HTTPS only to the Vultr VPS; Nginx forwards their requests through the VPN tunnel to the monolith on the hall's LAN, and the response returns by the same path. The database talks only to the monolith, and on-site staff use the local network.

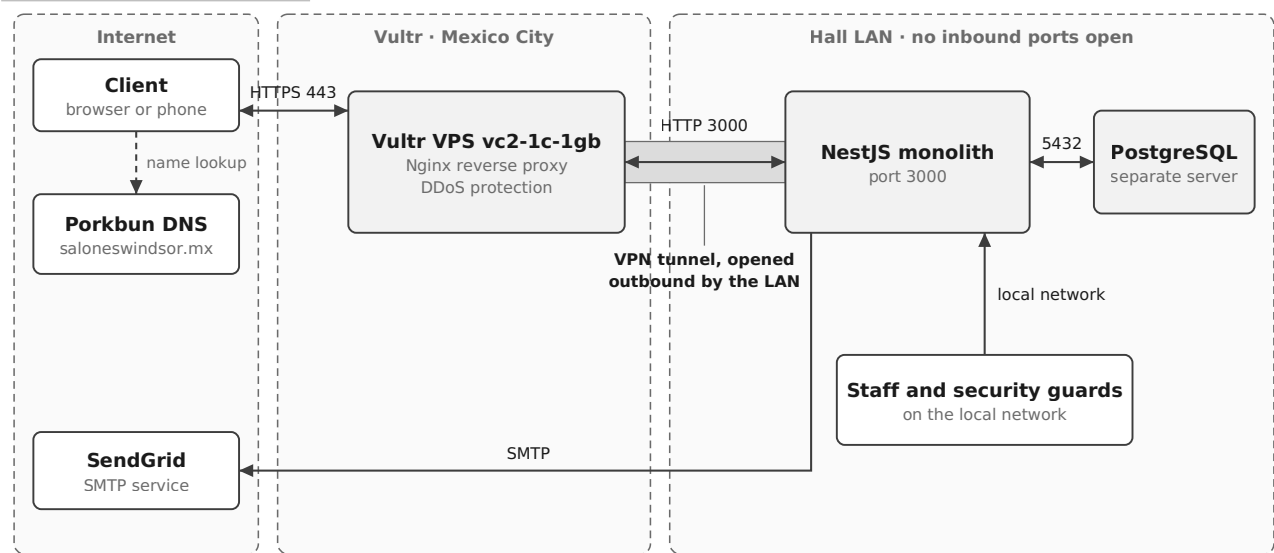


Figure 3. Deployment communication between the internet, the Vultr VPS, and the hall LAN

Figure 4 shows the messages exchanged when a Client books a hall (RF-CLI-004), and Table 11 describes each step. v0 does not define the calls between modules; this flow is derived from the requirements.

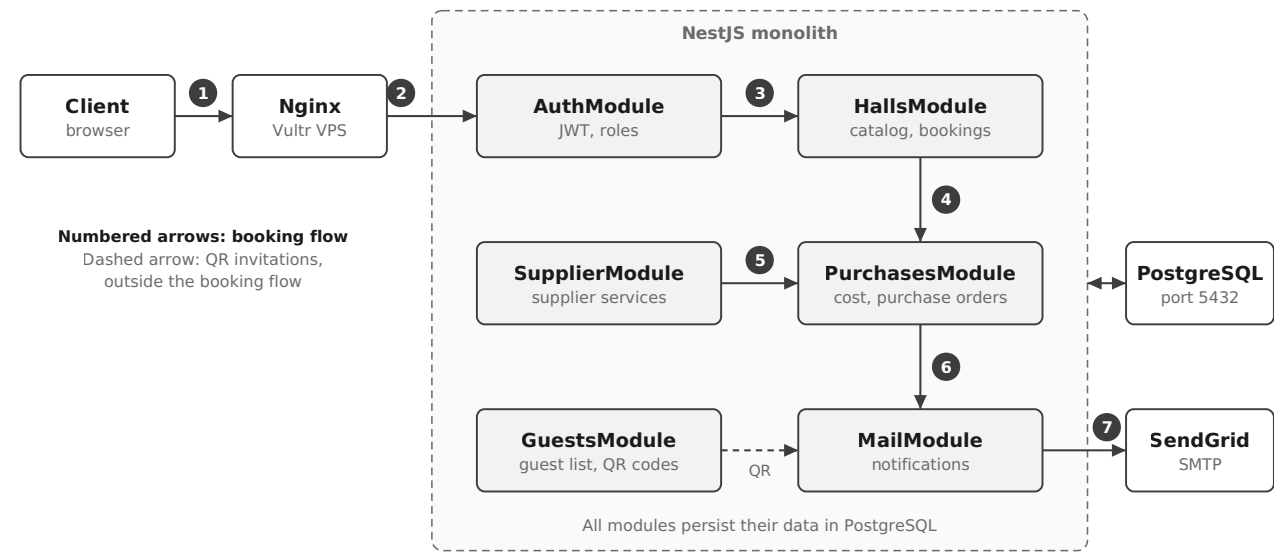


Figure 4. Module communication for the hall booking flow

Table 11. Booking flow messages

Step	From → To	Message	Requirements
1	Client → Nginx	HTTPS request from the booking view	RF-CLI-004
2	Nginx → AuthModule	Validate the JWT token and the Client role	RF-AUTH-002, RF-AUTH-003
3	AuthModule → HallsModule	Look up availability and record the booking without overlap	RF-SAL-002, RF-SAL-003
4	HallsModule → PurchasesModule	Calculate the total cost and create the purchase order	RF-COMP-001, RF-COMP-002
5	SupplierModule → PurchasesModule	Provide the prices of the supplier services linked to the booking	RF-PROV-002
6	PurchasesModule → MailModule	Request the booking confirmation email	RF-MAIL-001
7	MailModule → SendGrid	Deliver the email over SMTP	RF-MAIL-001

Deployment Alternatives

Twelve ways of giving Clients access from outside the LAN were compared. Six were discarded: a LAN-only kiosk and a public request form do not let the Client book online in real time; a separate booking portal and bidirectional database replication do not fit the schedule; exposing the LAN server directly is not secure; and a managed platform had no verified prices. **Table 12** scores the remaining six from 1 to 5, weighting schedule fit 40%, budget 20%, security 15%, availability 15%, and impact on the baseline 10%.

Table 12. Deployment alternatives evaluated

Alternative	Description	Annual cost	Effort (person-weeks)	Score
VPS proxy (selected)	Public Nginx proxy on a Vultr VPS with a VPN tunnel to the LAN	\$3,291 MXN	1	4.55
Cloudflare Tunnel	Tunnel from the LAN to Cloudflare, without a VPS	\$0 MXN	0.5	4.45
Full cloud	Application and database on two Vultr VPS	\$12,725 MXN	1	3.75
Managed database	Full cloud with a managed PostgreSQL database	\$10,751 MXN to \$27,206 MXN (unverified)	1	3.70
Hybrid direct	Application on a VPS using the LAN database over a VPN	\$7,460 MXN	2	3.10
Hybrid write/read	Primary database on the LAN; read replica and application on Vultr	\$11,848 MXN	3.5	2.90

The selected alternative fits in Sprints 3–4, leaves RNF-SEG-001 and RNF-SEG-002 unchanged, and keeps **Figure 2** valid. The hybrid write/read alternative reuses the same VPS and tunnel, so it remains the next step if the hall’s internet link proves unreliable.